On the irrelevance of irrelevant moment conditions*

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August 28, 2026

Abstract

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KEYWORDS: TBA

JEL classification codes: C12, C14.

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This paper contributes to the literature on inference in partially identified econometric models defined by a finite number of moment (in)equalities, i.e., inequalities and equalities. Consider an economic model with a parameter θ belonging to a parameter space Θ, whose main prediction is that the true value of θ, denoted by θ_0, satisfies a finite number of unconditional moment (in)equalities. We allow this model to be partially identified, i.e., the restrictions of the model do not necessarily restrict θ_0 to a single value, but rather they constrain it to belong to a certain set, called the identified set.

There is a vast and growing literature on inference in partially identified moment (in)equality models (see later in this section for a review). This paper investigates the effect that adding or subtracting moment conditions has on the asymptotic power of the inference in a partially identified moment (in)equality model. This analysis has important consequences in terms of the choice of the model used to conduct inference. To the best of our knowledge, this problem has not yet been fully explored by the existing literature.

Consider a researcher who assumes that θ_0 satisfies a partially identified moment (in)equality model denoted by $\mathcal{M}_L$. The subscript L in $\mathcal{M}_L$ indicates that this is a "long" model, with several moment (in)equalities. Let $\mathcal{M}_S$ denote a partially identified moment (in)equality model defined by an arbitrary subset of the moment (in)equalities in $\mathcal{M}_L$. The subscript S in $\mathcal{M}_S$ indicates that this is a "short" model, with less moment (in)equalities that $\mathcal{M}_L$. Since $\mathcal{M}_S \subset \mathcal{M}_L$, the researcher would also agree that θ_0 satisfies $\mathcal{M}_S$. Thus, the researcher could conduct inference on θ_0 based on either one of these moment (in)equality models. Suppose that the researcher is interested in the model whose inference is most powerful. Which one should the researcher choose?

From a population viewpoint, $\mathcal{M}_L$ is more or equally informative than $\mathcal{M}_S$. Intuition might then suggest that $\mathcal{M}_L$ results in relatively more informative inference, in the sense of statistical power of tests. This paper shows that this intuition can be incorrect. Specifically, we show that statistical inference based on $\mathcal{M}_L$ may or may not be more powerful than the one based on $\mathcal{M}_S$, even asymptotically. Whether this occurs or not depends on features of the inference problem that are ex-ante unknown to the researcher.

EXPLAIN THE INTUITION OF THE RESULT AROUND HERE.

LITERATURE REVIEW AROUND HERE. HERE ARE SOME RELATED PAPERS THAT TOUCH ON THIS PROBLEM:

- Tamer (2003) considers inference in a game with multiple equilibria, which takes the form of a partially identified moment (in)equality model. In the appendix, he investigates whether moment inequalities increase efficiency. Using an incorrect argument, he shows that additional moment inequalities cannot decrease asymptotic power.

- Moon and Schorfheide (2009) consider inference based on empirical likelihood in a point

identified moment (in)equality model. In this context, they investigate whether adding the moment inequalities to the pure moment equality model increases the power of the inference or not. They formally show that additional moment inequalities cannot decrease asymptotic power. Obviously, this is quite different from our result. The reason for the difference lies in the fact that their model is point-identified, so they can use empirical likelihood to conduct inference. In contrast, our model is partially identified, empirical likelihood is thus unavailable, and we need to use an inference method that is uniformly valid under partial identification.

- Galichon and Henry (2011) revisits the class of games with multiple equilibria. For computational reasons, their paper suggests eliminating moment (in)equalities that are slack (relative to the rest of the model). In particular, they define a "core determining" set of moment inequalities to be a set of moment inequalities that cannot be reduced any further (see Galichon and Henry (2011, Page 1275, Definition 7)). They only consider this set for computation at the population level, but they extend this to the sample in the application section. Their definition of core determining set is based on the value of the expectation of the moment inequality, i.e., $E_P[m_j(W,\theta)]$. Is this a good idea when one is faced with a random sample? Our paper shows that their definition of core determining set is not very good. Instead, one should consider Studentized versions, i.e., $E_P[m_j(W,\theta)]/\sqrt{Var_P[m_j(W,\theta)]}$.

- Pakes and Porter (2024) and Mbakop (2023) propose methods to eliminate redundant moment inequalities in discrete choice panel data models.

- Ponomarev and Semenova (2024, Remark 2) make a related point. But their point is about differences in variances, which is different. Ours is about critical values, even for the same asymptotic variance.

The remainder of this paper is organized as follows. In Section 1, we describe the general problem of inference in a partially identified moment (in)equality model using Generalized Moment Selection (GMS) in Andrews and Soares (2010). We focus on GMS because it is the leading inference method to conduct uniformly valid inference in the models. Section 2 is the main section of this paper. In this section, we propose two moment inequality models, $\mathcal{M}_S$ and $\mathcal{M}_L$, and use them to compare the local power of GMS inference. Section 3 illustrates the theoretical results using Monte Carlo simulations. Section 4 explores an idea to provide a concrete answer to the choice of the moment (in)equality model. Section 6 concludes the paper. All proofs and intermediate results are collected in the appendix of paper.

1 Inference in a partially identified moment (in)equality model

This paper considers inference on a parameter $\theta_0 \in \Theta$ that is partially identified by a moment (in)equality model. The moment (in)equality model is denoted by $\mathcal{M}$ and is given by:

$$\mathcal{M} \equiv \left\{ (\theta, P) \in \Theta \times \mathcal{P} : \left\{ \begin{array}{ll} E_P[m_j(W,\theta)] \geq 0, & j = 1, \ldots, p \\ E_P[m_j(W,\theta)] = 0, & j = p+1, \ldots, k \end{array} \right\} \right\},$$

where $m : \mathcal{W} \times \Theta \to \mathbb{R}^k$ is a known function, W is a random vector supported on $\mathcal{W}$ and distributed according to $P \in \mathcal{P} \equiv \mathcal{P}(a, M, \Psi)$, and $\mathcal{P}(a, M, \Psi)$ is the standard baseline probability space, defined next.

Definition 1.1 (Baseline probability space)**.** $\mathcal{P} \equiv \mathcal{P}(a, M, \Psi)$ is a set of distributions P such that, when paired with some $\theta \in \Theta$, the following conditions hold:

(a) $\{W_i : i \leq n\}$ are i.i.d. under P,

(b) $\sigma^2_{j,P}(\theta) = Var_P(m_j(W,\theta)) \in (0, \infty)$ for $j = 1, \ldots, k_2$,

(c) $Corr_P(m(W,\theta)) \in \Psi$,

(d) $E_P[|m_j(W,\theta)/\sigma_{j,P}(\theta)|^{2+\delta}] \leq M$,

where Ψ is a specified closed set of $k \times k$ correlation matrices, and M and a are fixed positive constants.

For any $P \in \mathcal{P}$, the identified set is denoted by $\Theta_I(P)$ and given by:

$$\Theta_I(P) \equiv \left\{ \theta \in \Theta : \left\{ \begin{array}{ll} E_P[m_j(W,\theta)] \geq 0, & j = 1, \ldots, p \\ E_P[m_j(W,\theta)] = 0, & j = p+1, \ldots, k \end{array} \right\} \right\}.$$

For a significance level of $\alpha \in (0,1)$ and for any $\theta \in \Theta$, we want to conduct the following hypothesis test:

$$H_0 : \theta_0 = \theta \quad \text{vs.} \quad H_1 : \theta_0 \neq \theta. \tag{1.1}$$

Our econometric model is not necessarily point identified, so we consider an inference procedure that allows for partial identification. We consider the GMS method in Andrews and Soares (2010), i.e., reject H_0 in (1.1) according to:

$$\phi_n(\theta, \alpha) \equiv \{Q_n(\theta) > c_n(\theta, 1-\alpha)\}, \tag{1.2}$$

where $\phi_n(\theta, \alpha)$ is an binary variable that indicates rejection, $Q_n(\theta)$ is the GMS test function, and $c_n(\theta, 1-\alpha)$ is the corresponding critical value. These are defined in detail in the next paragraph.

By definition, GMS confidence sets are the result of inverting the hypothesis test in (1.2), i.e.,

$$CS_n(1-\alpha) \;\equiv\; \{\theta \in \Theta : \phi_n(\theta, \alpha)\} = \{\theta \in \Theta : Q_n(\theta) > c_n(\theta, 1-\alpha)\}\,. \tag{1.3}$$

We now return to the various objects that define (1.2). The GMS criterion function is:

$$Q_n(\theta) \;\equiv\; S(\sqrt{n}\bar{m}_n(\theta), \hat{\Sigma}_n(\theta)), \tag{1.4}$$

where $\bar{m}_n(\theta) \equiv n^{-1}\sum_{i=1}^n m(W_i, \theta)$ and $\hat{\Sigma}_n(\theta) \equiv n^{-1}\sum_{i=1}^n (m(W_i,\theta) - \bar{m}_n(\theta))(m(W_i,\theta) - \bar{m}_n(\theta))'$, and S is the test function that satisfies Assumptions 1 and 2, presented next. The assumption makes use of the following notation: $\mathbb{R}_{[+\infty]} \equiv \mathbb{R} \cup \{\infty\}$, and $\mathbb{R}_{[\pm\infty]} \equiv \mathbb{R} \cup \{\infty\} \cup \{-\infty\}$.

Assumption 1 (Test function S)**.** $S(m, \Sigma) : \mathbb{R}^k_{[\pm\infty]} \times \mathbb{R}^{k\times k} \to \mathbb{R}_{[+\infty]}$ satisfies

(a) $S((m_1, m_2), \Sigma)$ is nonincreasing in m_1 for all $(m_1, m_2) \in \mathbb{R}^p_{[+\infty]} \times \mathbb{R}^{k-p}$ and variance matrices $\Sigma \in \mathbb{R}^{k\times k}$,

(b) $S(m, \Sigma) = S(Dm, D\Sigma D)$ for all $m \in \mathbb{R}^k$, $\Sigma \in \mathbb{R}^{k\times k}$, and positive definite diagonal matrix $D \in \mathbb{R}^{k\times k}$,

(c) $S(m, \Omega) \geq 0$ for all $m \in \mathbb{R}^k$ and $\Omega \in \Psi$,

(d) $S((m_1, m_2), \Omega)$ is continuous at all $m_1 \in \mathbb{R}^p_{[+\infty]}$, $m_2 \in \mathbb{R}^{k-p}$, and $\Omega \in \Psi$.

(e) $S(m, \Omega) > 0$ if and only if $m_j < 0$ for some $j = 1, \ldots, p$, or $m_j \neq 0$ for some $j = p+1, \ldots, k$, where $m = (m_1, \ldots, m_k)$ and $\Omega \in \Psi$.

Assumption 2 (Limiting distribution)**.** For all $h \in \prod_{j=1}^p [-r_j, \infty] \times \prod_{j=p+1}^v [-r_j, r_j]$, all $\Omega \in \Psi$, and $Z \sim N(\mathbf{0}_k, \Omega)$, the cumulative distribution function of $S(Z + (h_1, \mathbf{0}_{k-p}), \Omega)$ at $x \in \mathbb{R}$ is

(a) continuous for $x > 0$,

(b) strictly increasing for $x > 0$ unless $p = k$ and $h_1 = \infty^p$, and

(c) less than or equal to $1/2$ at $x = 0$ when $v \geq 1$ or when $v = 0$ and $h_j = 0$ for some $j = 1, \ldots, p$.

Remark 1.1. Assumption 1 is the union of Assumptions 1 and 3 in Andrews and Soares (2010). Assumption 2 is Assumption 2 in Andrews and Soares (2010).

Assumptions 1-2 are satisfied by all test functions in Andrews and Soares (2010). These include the "maximum" test function:

$$S_{\max}(x, \Sigma) \;=\; \max\{(-x_j/\Sigma_{(j,j)} : j = 1, \ldots, p)', (|x_j|/\Sigma_{(j,j)} : j = p+1, \ldots, k)'\} \tag{1.5}$$

and the "modified method of moments" test function:

$$S_{\text{mmm}}(x,\Sigma) \;=\; \sum_{j=1}^{p}[x_j/\Sigma_{(j,j)}]_-^2 + \sum_{j=p+1}^{k}[x_j/\Sigma_{(j,j)}]^2. \tag{1.6}$$

Both (1.5) and (1.6) presume that the variance-covariance matrix Σ has positive diagonal, and one could avoid this problem by replacing $x_j/\Sigma_{(j,j)}$ with $x_j/\Sigma_{(j,j)}$ if $\Sigma_{(j,j)} > 0$ and 0 if $\Sigma_{(j,j)} = 0$. ■

In turn, $\hat{c}_n(\theta)$ is the (conditional) $(1-\alpha)$-quantile of

$$Q_n^*(\theta) \;\equiv\; S(\, v_n^*(\theta) + \varphi(\kappa_n^{-1}\sqrt{n}\hat{D}_n(\theta)^{-1/2}\bar{m}_n(\theta), \hat{\Omega}_n(\theta))\;,\; \hat{\Omega}_n(\theta)\,), \tag{1.7}$$

where $\hat{D}_n(\theta) \equiv diag(\hat{\Sigma}_n(\theta))$, $\hat{\Omega}_n(\theta) \equiv \hat{D}_n(\theta)^{-1/2}\hat{\Sigma}_n(\theta)\hat{D}_n(\theta)^{-1/2}$, $v_n^*(\theta) \equiv n^{-1/2}\sum_{i=1}^n \hat{D}_n(\theta)^{-1/2}(m(W_i,\theta) - \bar{m}_n(\theta))\zeta_i$, $\{\zeta_i : i \leq n\}$ is an i.i.d. sample with $\zeta_i \sim N(0,1)$ that is independent of the data, $\{\kappa_n : n \geq 1\}$ is a tuning parameter sequence that satisfies $\kappa_n \to \infty$ and $\kappa_n/\sqrt{n} \to 0$, and φ is the GMS function that satisfies Assumption 3, defined in the next paragraph. Again, (1.7) presumes that the variance-covariance matrix $\hat{\Sigma}_n(\theta)$ has positive diagonal, and one could avoid this problem by replacing $v_n^*(\theta)$, $\hat{D}_n(\theta)^{-1/2}$, and $\hat{\Omega}_n(\theta)$ with $v_n^*(\theta)I\{\min \hat{D}_n(\theta) > 0\}$, $\hat{D}_n(\theta)^{-1/2}I\{\min \hat{D}_n(\theta) > 0\}$, and $\hat{\Omega}_n(\theta)I\{\min \hat{D}_n(\theta) > 0\}$, respectively.

Assumption 3 (GMS function φ)**.** The function $\varphi(\xi,\Omega) : \mathbb{R}^p_{[\pm\infty]} \times \mathbb{R}^{k\times k} \to \mathbb{R}^k_{[+\infty]}$ is defined by $\varphi(\xi,\Omega) = \{\varphi_j(\xi, \mathbf{0}_{k-p}, \Omega) : j = 1,\ldots,k\}$, where $\varphi_j(\xi, \mathbf{0}_{k-p}, \Omega) \in \mathbb{R}$ satisfies the following conditions:

(a) For some set $C \subseteq \mathbb{R}$ with $0 \in C$, $\varphi_j(\xi,\Omega)$ is continuous at all $(\xi,\Omega) \in \mathbb{R}^p_{[+\infty]} \times \mathbb{R}^{k-p}_{[\pm\infty]} \times \Psi$ with $\xi_j \in C$, where $\xi = (\xi_1,\ldots,\xi_k)'$ for $j = 1,\ldots,p$.

(b) $\varphi_j(\xi,\Omega) = 0$ for all $(\xi,\Omega) \in \mathbb{R}^p_{[+\infty]} \times \mathbb{R}^{k-p}_{[\pm\infty]} \times \Psi$ with $\xi_j = 0$, where $\xi = (\xi_1,\ldots,\xi_k)'$ for $j = 1,\ldots,p$.

(c) $\varphi_j(\xi,\Omega) = 0$ for all $j = p+1,\ldots,k$ for all $(\xi,\Omega) \in \mathbb{R}^p_{[+\infty]} \times \mathbb{R}^{k-p}_{[\pm\infty]} \times \Psi$.

(d) $\kappa_n \to \infty$ and $\kappa_n/\sqrt{n} \to \infty$.

Remark 1.2. Assumption 3 is the union of Assumptions GMS1, GMS3, and GMS6 in Andrews and Soares (2010).

Assumption 3 is satisfied by all GMS functions in Andrews and Soares (2010). These include the following examples:

1. For $C = \mathbb{R}\backslash\{1\}$, $\varphi_j(\xi,\Omega) = \infty \times I\{\xi_j > 1\}$,

2. $\varphi_j(\xi,\Omega) = \xi_j$,

3. $\varphi_j(\xi, \Omega) = [\xi_j]_+$,

4. For $C = \mathbb{R}\backslash\{a_L, a_H\}$ for $\varphi_j(\xi, \Omega) = \phi(\xi) \times I\{a_L \leq \xi_j \leq a_H\} + \infty \times I\{a_H \leq \xi_j\}$, where $0 < a_L < a_H < \infty$ and $\phi : \mathbb{R}_{[+\infty]} \to [0, \infty]$ is an arbitrary nondecreasing function.

■

Under Assumptions 1, 2, and 3, Andrews and Soares (2010, Theorem 1) shows that the GMS tests provide uniform asymptotic size control, i.e.,

$$\limsup_{n\to\infty} \sup_{P\in\mathcal{P}} \sup_{\theta\in\Theta_I(P)} E_P[\phi_n(\theta, \alpha)] \quad \leq \quad \alpha.$$

This paper focuses on the power properties of these tests, i.e., on the limiting behavior of $E_{P_n}[\phi_n(\theta_n, \alpha)]$ along sequences of $\{(\theta_n, P_n) \in \Theta \times \mathcal{P}\}$ that do not satisfy H_0 in (1.1).

2 A simple setup

The remainder of this paper focuses on a simple example with two moment inequalities and zero moment equalities. The researcher assumes that true parameter value $\theta_0 \in \Theta \equiv [-B, B]$ satisfies the following partially identified moment inequality models:

$$\begin{aligned} \mathcal{M}_S &= \{(\theta, P) \in \Theta \times \mathcal{P} : E_P[W_1 - \theta] \geq 0\}, \\ \mathcal{M}_L &= \{(\theta, P) \in \Theta \times \mathcal{P} : \{E_P[W_1 - \theta] \geq 0 \ \cap \ E_P[W_2 - \theta] \geq 0\}\}, \end{aligned} \tag{2.1}$$

where $W = (W_1, W_2) \sim P \in \mathcal{P}$. Note that $\mathcal{M}_L$ includes the moment inequality in $\mathcal{M}_S$, plus an additional one, i.e., from a population viewpoint, $\mathcal{M}_L$ is more or equally informative than $\mathcal{M}_S$. Let us denote $\mu_{1,P} \equiv E_P[W_1]$ and $\mu_{2,P} \equiv E_P[W_2]$. The corresponding identified sets are:

$$\begin{aligned} \Theta_{I,S}(P) &= \{\theta \in [-B, B] : \theta \leq \mu_{1,P}\} \\ \Theta_{I,L}(P) &= \{\theta \in [-B, B] : \theta \leq \min\{\mu_{1,P}, \mu_{2,P}\}\}. \end{aligned}$$

We can represent a situation of irrelevance of the second moment inequality when $\mu_{1,P} \leq \mu_{2,P}$.

Remark 2.1. The conclusions in this section extend beyond the two-moment-inequality model to general moment (in)equality models with $p \geq 2$ moment inequalities and $k - p$ moment equalities, provided the additional moment conditions involve random variables independent of (W_1, W_2). If the $p - 2$ extra moment inequalities are sufficiently slack, they do not affect the test statistic or critical value. The non-slack moment inequalities and the $k - p$ moment equalities increase both the test statistic and the critical value, but do not qualitatively affect the rejection decisions of the GMS test. ■

For any $\theta \in \Theta$ and for a significance level of $\alpha \in (0,1)$, our goal is to conduct the hypothesis test in (1.2) test using either $\mathcal{M}_S$ or $\mathcal{M}_L$. Both of these econometric models are partially identified, and so we implement inference with the GMS method with the following test functions:

$$\begin{aligned} Q_{n,S}(\theta) &= S(\sqrt{n}(\bar{W}_1 - \theta)/\hat{\sigma}_1, 1) \\ Q_{n,L}(\theta) &= S((\sqrt{n}(\bar{W}_1 - \theta)/\hat{\sigma}_1, \sqrt{n}(\bar{W}_2 - \theta)/\hat{\sigma}_2)', [1, \hat{\rho}; \hat{\rho}, 1]) \end{aligned}$$

respectively, where $\bar{W}_s = n^{-1}\sum_{i=1}^n W_{s,i}$ and $\hat{\sigma}_s^2 = n^{-1}\sum_{i=1}^n (W_{s,i} - \bar{W}_s)^2$ for $s = 1, 2$, and $\hat{\rho} = n^{-1}\sum_{i=1}^n (W_{1,i} - \bar{W}_1)(W_{2,i} - \bar{W}_2)/(\hat{\sigma}_1\hat{\sigma}_2)$. (Once again, this notation assumes that $\hat{\sigma}_1\hat{\sigma}_2 > 0$, but could be replaced by $\hat{\rho} = n^{-1}\sum_{i=1}^n (W_{1,i} - \bar{W}_1)(W_{2,i} - \bar{W}_2)/(\hat{\sigma}_1\hat{\sigma}_2)$ if $\hat{\sigma}_1\hat{\sigma}_2 > 0$ and $\hat{\rho} = 0$ if $\hat{\sigma}_1\hat{\sigma}_2 = 0$. Under our assumptions, we have that $\hat{\sigma}_1\hat{\sigma}_2 > 0$ w.p.a.1). The GMS critical values in (1.7) for $\mathcal{M}_S$ and $\mathcal{M}_L$ are the (conditional) $(1-\alpha)$-quantiles of:

$$\begin{aligned} Q^*_{n,S}(\theta) &= S(v^*_{n,1} + \varphi_1(\kappa_n^{-1}\sqrt{n}(\bar{W}_1 - \theta)/\hat{\sigma}_1, 1)) \\ Q^*_{n,L}(\theta) &= S\left(\begin{pmatrix} v^*_{n,1} + \varphi_1(\kappa_n^{-1}\sqrt{n}(\bar{W}_1 - \theta)/\hat{\sigma}_1, \kappa_n^{-1}\sqrt{n}(\bar{W}_2 - \theta)/\hat{\sigma}_2, [1, \hat{\rho}; \hat{\rho}, 1]), \\ v^*_{n,2} + \varphi_2(\kappa_n^{-1}\sqrt{n}(\bar{W}_1 - \theta)/\hat{\sigma}_1, \kappa_n^{-1}\sqrt{n}(\bar{W}_2 - \theta)/\hat{\sigma}_2, [1, \hat{\rho}; \hat{\rho}, 1]) \end{pmatrix}, [1, \hat{\rho}; \hat{\rho}, 1] \right), \end{aligned}$$

respectively, where $v^*_{n,s} = n^{-1/2}\sum_{i=1}^n (W_{s,i} - \bar{W}_s)\zeta_i/\hat{\sigma}_s$ for $s = 1, 2$, $\{\zeta_i : i \leq n\}$ is an i.i.d. sample with $\zeta_i \sim N(0,1)$ that is independent of the data. (Once again, the notation assumes that $\hat{\sigma}_1\hat{\sigma}_2 > 0$, but this could be fixed by redefining these as zero whenever the corresponding sample variance is zero. Under our assumptions, we have that $\hat{\sigma}_1\hat{\sigma}_2 > 0$ w.p.a.1).

To study statistical power, we consider a sequence of $\{(\theta_n, P_n) \in \Theta \times \mathcal{P}\}$ that does not satisfy H_0 in (1.1), i.e., $\theta_n > \min\{\mu_{1,P_n}, \mu_{2,P_n}\}$. If irrelevance is maintained under H_1, then, $\mu_{1,P_n} \leq \mu_{2,P_n}$, and so $\theta_n > \mu_{1,P_n} = \min\{\mu_{1,P_n}, \mu_{2,P_n}\}$.

Without significant loss of generality, we consider the convergence of the following sequences in the extended real line:

$$\begin{aligned} h_1 &\equiv \lim \sqrt{n}(\theta_n - \mu_{1,P_n})/\sigma_{1,P_n} \\ \eta_1 &\equiv \lim \kappa_n^{-1}\sqrt{n}(\theta_n - \mu_{1,P_n})/\sigma_{1,P_n} \\ h_2 &\equiv \lim \sqrt{n}(\theta_n - \mu_{2,P_n})/\sigma_{2,P_n} \\ \eta_2 &\equiv \lim \kappa_n^{-1}\sqrt{n}(\theta_n - \mu_{2,P_n})/\sigma_{2,P_n} \\ \rho &\equiv \lim \rho_{P_n}. \end{aligned} \tag{2.2}$$

Note that $\kappa_n \to \infty$ implies that $\eta_s \leq h_s$ and $\eta_s = 0$ if $h_s \in \mathbb{R}$ for $s = 1, 2$. The convergence in (2.2) is without loss of generality, as it occurs by considering suitable subsequences $\{k_n : n \in \mathbb{N}\}$.

We consider sequences of $\{(\theta_n, P_n) \in \Theta \times \mathcal{P}\}$ that are local alternative hypotheses, i.e., that do not satisfy H_0 in (1.1). By definition of (local) alternatives, at least one of the moment inequalities

in $\mathcal{M}_L$ cannot be slack, i.e., either $(h_1, \eta_1) \geq (0,0)$, or $(h_2, \eta_2) \geq (0,0)$, or both. The parameters $(h_1, \eta_1, h_2, \eta_2)$ provide a measure of the extent of violation of the moment inequalities under consideration. Even if $\theta_n > \mu_{1,P_n}$, the presence of σ_{1,P_n} and σ_{2,P_n} can generate any possibility for these parameters. In words, a small violation of the null hypothesis that is very precisely measured can be easier to detect than a big violation that is imprecise.

Theorem 2.1 (Power for general sequences)**.** Consider GMS inference applied to (2.1), where S satisfies Assumption 1-2 and φ satisfies Assumption 3. Assume that the sequence $\{(P_n, \theta_n) \in \mathcal{P} \times \Theta\}_{n \geq 1}$ with $\rho_{P_n} \to \rho$, $\sigma_{j,P_n} \to \sigma_j > 0$ for $j = 1, 2$, $\sqrt{n}(\theta_n - \mu_{j,n})/\sigma_{j,n} \to h_j \in \mathbb{R}_{[+\infty]}$ for $j = 1, 2$, and $\kappa_n^{-1}\sqrt{n}(\theta_n - \mu_{j,n})/\sigma_{j,n} \to \eta_j \in C$ for $j = 1, 2$, where C and κ_n are as specified in Assumption 3. Furthermore, consider the CDFs:

$$
\begin{aligned}
F_S(x) &= P\left(S\left(\xi_1 + \varphi_1\left(-\eta_1, 1\right), 1\right) \leq x\right) \\
F_L(x) &= P\left(S\left(\xi_1 + \varphi_1((-\eta_1, -\eta_2), \Omega), \xi_2 + \varphi_2((-\eta_2, -\eta_1), \Omega), \Omega\right) \leq x\right),
\end{aligned}
$$

with $(\xi_1, \xi_2) \sim N(\mathbf{0}_2, \Omega)$ and $\Omega = [1, \rho; \rho, 1]$. Finally, we assume that F_S and F_L are strictly increasing at their $(1-\alpha)$-quantile, denoted by $c_S(\eta_1)$ and $c_L(\eta_1, \eta_2, \rho)$, respectively. Then,

1. $\lim_{n\to\infty} P_n(Q_{n,S}(\theta_n) > \hat{c}_{n,S}(\theta_n)) = P(S(\xi_1 + h_1, 1) > c_S(\eta_1))$.
2. $\lim_{n\to\infty} P_n(Q_{n,L}(\theta_n) > \hat{c}_{n,L}(\theta_n)) = P(S((\xi_1 + h_1, \xi_2 + h_2)', [1, \rho; \rho, 1]) > c_L(\eta_1, \eta_2, \rho))$.

Proof. This is a corollary of Lemmas A.1, A.2, A.3. ■

Under the assumptions in Theorem 2.3, we can compare the power of the GMS method between the two models by comparing the expressions $P(S(\xi_1 + h_1, 1) > c_S(\eta_1))$ and $P(S((\xi_1 + h_1, \xi_2 + h_2)', [1, \rho; \rho, 1]) > c_L(\eta_1, \eta_2, \rho))$. To simplify the comparison, we consider a particularly simple case.

Theorem 2.2 (Power for all local sequences)**.** Consider GMS inference applied to (2.1), where S satisfies Assumption 1-2 and φ satisfies Assumption 3. Assume that the sequence $\{(P_n, \theta_n) \in \mathcal{P} \times \Theta\}_{n \geq 1}$ with $\rho_{P_n} \to \rho$, $\sigma_{j,P_n} \to \sigma_j > 0$ for $j = 1, 2$, $\sqrt{n}(\theta_n - \mu_{j,n})/\sigma_{j,n} \to h_j \in \mathbb{R}$ for $j = 1, 2$. Then,

1. $\lim_{n\to\infty} P_n(Q_{n,S}(\theta_n) > \hat{c}_{n,S}(\theta_n)) = P(S(\xi_1 + h_1, 1) > c_S)$, where c_S is the $(1-\alpha)$-quantile of $F_S(x) = P(S(\xi_1, 1) \leq x)$.
2. $\lim_{n\to\infty} P_n(Q_{n,L}(\theta_n) > \hat{c}_{n,L}(\theta_n)) = P(S((\xi_1 + h_1, \xi_2 + h_2)', [1, \rho; \rho, 1]) > c_L(\rho))$, where $c_L(\rho)$ is the $(1-\alpha)$-quantile of $F_L(x) = P(S((\xi_1, \xi_2)', [1, \rho; \rho, 1]) \leq x)$.

Proof. This is a corollary of Lemmas A.1, A.2, A.3. ■

To make the main point of this paper, we make the following assumption.

Definition 2.1 (Removing a coordinate)**.** For any $j = 1, \ldots, p$, $h_1 \in \mathbb{R}^p_{[\pm\infty]} = [-\infty, \infty]^p$, and $\Omega \in \Psi$, define $S_{-j}(Z + (h_1, \mathbf{0}_{k-p}), \Omega)$ as the random variable generated by $S(Z + (h_1, \mathbf{0}_{k-p}), \Omega)$ but with the j-th moment inequality removed, i.e.,

$$\tilde{S}\left(Z_{-j} + (h_{1,-j}, \mathbf{0}_{k-p}), \Omega_{-j}\right),$$

with $\tilde{S} : \mathbb{R}^{k-1} \times \mathbb{R}^{k-1 \times k-1} \to \mathbb{R}$, and

$$\begin{aligned} Z_{-j} &= \{Z_a : a = \{1, \ldots, p\} \setminus j\} \\ h_{1,-j} &= \{h_{1,a} : a = \{1, \ldots, p\} \setminus j\} \\ \Omega_{-j} &= \{\Omega[a, b] : a, b = \{1, \ldots, p\} \setminus j\}. \end{aligned}$$

Also, let $c_{-j}(\Omega)$ are the $(1-\alpha)$-quantile of $\tilde{S}\left(Z_{-j} + (h_{1,-j}, \mathbf{0}_{k-p}), \Omega_{-j}\right)$, i.e.,

$$c_{-j}(\Omega) = \inf_x \left\{ P\left(\tilde{S}\left(Z_{-j} + (h_{1,-j}, \mathbf{0}_{k-p}), \Omega_{-j}\right) \leq x\right) \geq 1 - \alpha \right\}.$$

Assumption 4 (Removal of coordinates via $h_1 \to -\infty$)**.** For any $j = 1, \ldots, p$, $h_1 \in \mathbb{R}^p_{[\pm\infty]}$, and $\Omega \in \Psi_0$, we have

$$S(Z + (h_1^{-j}, \mathbf{0}_{k-p}), \Omega) = \tilde{S}\left(Z_{-j} + (h_{1,-j}, \mathbf{0}_{k-p}), \Omega_{-j}\right),$$

where

$$h_1^{-j} = \{h_{1,a} I\{a \neq j\} - \infty I\{a = j\} : a = 1, \ldots, p\}.$$

Remark 2.2. Assumption 4 is true for $S_{\max}$, S_1, and S_3 with $\Psi_0 = \Psi$. It is also true for S_2, provided that Ψ_0 is diagonal (in that case, $S_2 = S_3$). ■

Assumption 5 (Limiting distribution for local alternatives)**.** We have

(a) For any $x \in (0, \infty)$, $j = 1, \ldots, p$, $h_{1,a} \in (0, \infty)$ for $a = \{1, \ldots, p\} \setminus j$, and $\Omega \in \Psi$,

$$P\left(S(Z + \left(h_1^{j,0}, \mathbf{0}_{k-p}\right), \Omega) \leq x\right) > P\left(S(Z + \left(h_1^{-j}, \mathbf{0}_{k-p}\right), \Omega) \leq x\right)$$

where $h_1^{j,0} = \{h_{1,a} I\{a \neq j\} : a = 1, \ldots, p\}$ and $h_1^{-j} = \{h_{1,a} I\{a \neq j\} - \infty I\{a = j\} : a = 1, \ldots, p\}$.

In words: all moment inequalities indicate a violation of H_0. If the jth moment inequality has no signal of violation, it dilutes the remaining signals. Deleting the jth coordinate strictly increases the rejection rate.

Assumption 1(a) gives a weak inequality. This adds strict when we go to the extremes. It is a very mild assumption.

(b) For any $j = 1, \ldots, p$, $\lambda \in (0, \infty)$, and $\bar{h}_1 = \{\lambda : a = 1, \ldots, p\}$, and $\Omega \in \Psi_r$ (restricted to avoid

exact duplication of signals),

$$P\left(S(Z+\left(\bar{h}_1^{-j},\mathbf{0}_{k-p}\right),\Omega)\leq c_{-j}\left(\Omega\right)\right)>P\left(S(Z+\left(\bar{h}_1,\mathbf{0}_{k-p}\right),\Omega)\leq c\left(\Omega\right)\right)$$

where $\bar{h}_1^{-j}=\{\lambda I\{a\neq j\}-\infty I\{a=j\}:a=1,\ldots,p\}$, $c\left(\Omega\right)$ is the $(1-\alpha)$-quantile of $S(Z,\Omega)$.

In words: all moment inequalities have the same signal. If we remove a coordinate, that decreases the power. Having many equal signals of rejection (all equal to λ) is strictly better than having less of them.

(c) For any $x\in(0,\infty)$, $h_1\in(0,\infty)^p$, and $\Omega\in\Psi$, $P\left(S(Z+(h_1,\mathbf{0}_{k-p}),\Omega)\leq x\right)$ is a continuous function of h_1.

For our purposes, we can specialize Assumption 5 for the case with $p=j=2$.

Assumption 6 (Limiting distribution for local alternatives $p=j=2$)**.** We have

(a) For any $x\in(0,\infty)$, $h_1\in(0,\infty)$, and $\rho\in[-1,1]$,

$$\begin{aligned}P\left(S((Z_1+h_1,Z_2)',[1,\rho;\rho,1])\leq x\right)&>P\left(S((Z_1+h_1,-\infty)',[1,\rho;\rho,1])\leq x\right)\\&=P\left(S(Z_1+h_1,1)\leq x\right),\end{aligned}$$

where the last equality uses Assumption 4.

(b) For any $\lambda\in(0,\infty)$ and $\rho\in[-1,1)$,

$$\begin{aligned}P\left(S(Z_1+\lambda,1)\leq c_S\right)&=P\left(S((Z_1+\lambda,Z_2-\infty)',[1,\rho;\rho,1])\leq c_S\right)\\&>P\left(S((Z_1+\lambda,Z_2+\lambda)',[1,\rho;\rho,1])\leq c_L(\rho)\right),\end{aligned}$$

where the first equality uses Assumption 4, $c_S=\inf\{x:P\left(S(Z_1,1)\leq x\right)\geq 1-\alpha\}$ and $c_L(\rho)=\inf\left\{x:P\left(S((Z_1,Z_2)',[1,\rho;\rho,1])\leq x\right)\geq 1-\alpha\right\}$, and the last one that $c_S\leq c_L(\rho)$.

(c) For any $x\in(0,\infty)$, $P\left(S(Z_1+h_1,1)\leq x\right)$ and $P\left(S((Z_1+h_1,Z_2+h_2)',[1,\rho;\rho,1])\leq x\right)$ are continuous functions of (h_1,h_2).

Theorem 2.3 (Power comparison is ambiguous)**.** Consider GMS inference applied to (2.1), where S satisfies Assumption 1, 2, 4, and 6 and φ satisfies Assumption 3. Assume that the sequence $\{(P_n,\theta_n)\in\mathcal{P}\times\Theta\}_{n\geq 1}$ with $\rho_{P_n}\to\rho$, $\sigma_{j,P_n}\to\sigma_j>0$ for $j=1,2$, $\sqrt{n}(\theta_n-\mu_{j,n})/\sigma_{j,n}\to h_j\in\mathbb{R}_+$ for $j=1,2$, and $[1,\rho;\rho,1]\in\Psi_0$ with Ψ_0 as in Assumption 4. Then,

(a) There is $h_{2,A}\in(0,h_1)$ *small* enough s.t. $\Pi_L\left(h_1,h_{2,A},\rho\right)<\Pi_S\left(h_1\right)$.

(b) There is $h_{2,B}\in(0,h_1)$ *large* enough s.t. $\Pi_S\left(h_1\right)<\Pi_L\left(h_1,h_{2,B},\rho\right)$.

Proof. First, we note that our assumptions imply $c_S \leq c_L(\rho)$. To see why, consider the following argument:

$$1-\alpha \overset{(1)}{\leq} P\left(S((Z_1,Z_2)', [1,\rho;\rho,1]) \leq c_L(\rho)\right) \overset{(2)}{\leq} P\left(S(Z_1,1) \leq c_L(\rho)\right), \tag{2.3}$$

where (1) holds by definition of $c_L(\rho)$, and (2) by Assumptions 1(a) and 4. By definition of c_S and (2.3), the desired result follows.

Second, consider the following argument.

$$\begin{aligned} P\left(S((\xi_1+h_1,\xi_2)', [1,\rho;\rho,1]) \leq c_L(\rho)\right) &\overset{(1)}{\geq} P\left(S((\xi_1+h_1,\xi_2)', [1,\rho;\rho,1]) \leq c_S\right) \\ &\overset{(2)}{>} P\left(S(\xi_1+h_1,1) \leq c_S\right) \\ &\overset{(3)}{>} P\left(S((\xi_1+h_1,\xi_2+h_1)', [1,\rho;\rho,1]) \leq c_L(\rho)\right), \end{aligned} \tag{2.4}$$

where (1) holds by $c_S \leq c_L(\rho)$, (2) by Assumption 6(a), and (3) by Assumption 6(b). We are now ready to prove both results.

Part (a). By inequalities (1)-(2) in (2.4), we have that

$$P\left(S((\xi_1+h_1,\xi_2)', [1,\rho;\rho,1]) > c_L(\rho)\right) < P\left(S(\xi_1+h_1,1) > c_S\right) = \Pi_S(h_1). \tag{2.5}$$

By (2.5) and Assumption 6(c), we can find $h_{2,A} > 0$ and sufficiently close to zero s.t.

$$\Pi_L(h_1, h_{2,A}, \rho) = P\left(S((\xi_1+h_1,\xi_2+h_{2,A})', [1,\rho;\rho,1]) > c_L(\rho)\right) < \Pi_S(h_1),$$

as desired. Note that $h_{2,A} < h_1$, as if $h_{2,A} \geq h_1$ would imply a contradiction with inequality (3) in (2.4).

Part (b). By inequality (3) in (2.4) implies

$$\Pi_S(h_1) < P\left(S((\xi_1+h_1,\xi_2+h_1)', [1,\rho;\rho,1]) > c_L(\rho)\right).$$

By (2.5) and Assumption 6(c), we can find $h_{2,B} < h_1$ and sufficiently close to h_1 s.t.

$$\Pi_S(h_1) < P\left(S((\xi_1+h_1,\xi_2+h_{2,B})', [1,\rho;\rho,1]) > c_L(\rho)\right) = \Pi_L(h_1, h_{2,B}, \rho),$$

as desired. Note that $h_{2,B} > 0$, as if $h_{2,B} = 0$ would imply a contradiction with inequalities (1)-(2) in (2.4). ■

Assumption 7 (Monotonicity of the long-model limiting distribution)**.** Fix $h_1 > 0$ and $\rho \in (-1, 1)$, and let

$$\Omega_\rho \;=\; [1, \rho; \rho, 1].$$

For every $h_2 \in \mathbb{R}$, define

$$F_L(x; h_1, h_2, \rho) \;=\; P\left(S\left((\xi_1 + h_1, \xi_2 + h_2)', \Omega_\rho\right) \le x\right),$$

where $(\xi_1, \xi_2)' \sim N(0, \Omega_\rho)$. The following conditions hold.

(a) The function

$$h_2 \mapsto F_L(c_L(\rho); h_1, h_2, \rho)$$

is continuous and strictly decreasing.

(b) As the second moment inequality becomes infinitely violated, the long statistic diverges:

$$\lim_{h_2 \to \infty} F_L(x; h_1, h_2, \rho) \;=\; 0 \quad \text{for every finite } x \in \mathbb{R}.$$

In words, the statistic reacts to arbitrarily large signal from the second moment inequality.

(c) The long-model power is below the short-model power when the second local signal is sufficiently small:

$$\lim_{h_2 \to -\infty} \Pi_L(h_1, h_2, \rho) \;<\; \Pi_S(h_1).$$

In words, the signal from the first moment inequality is eventually diluted when the second moment inequality becomes sufficiently slack.

Theorem 2.4 (Unique power cutoff)**.** Consider GMS inference applied to (2.1), where S satisfies Assumptions 1–2 and φ satisfies Assumption 3. Fix $h_1 > 0$ and $\rho \in (-1, 1)$. Suppose Assumption 7 holds. Also suppose that

$$\Pi_S(h_1) < 1.$$

Then there exists a unique cutoff $\bar{h}_2 \in \mathbb{R}$ such that

$$\Pi_L(h_1, \bar{h}_2, \rho) \;=\; \Pi_S(h_1).$$

Moreover,

$$\Pi_L(h_1, h_2, \rho) \;<\; \Pi_S(h_1) \quad \text{for all } h_2 < \bar{h}_2,$$

and

$$\Pi_S(h_1) \;<\; \Pi_L(h_1, h_2, \rho) \quad \text{for all } h_2 > \bar{h}_2.$$

Proof. By definition of the long-model limiting power,

$$\Pi_L(h_1, h_2, \rho) \;=\; P\left(S\left((\xi_1 + h_1, \xi_2 + h_2)', \Omega_\rho\right) > c_L(\rho)\right).$$

Equivalently,

$$\Pi_L(h_1, h_2, \rho) \;=\; 1 - F_L(c_L(\rho); h_1, h_2, \rho).$$

By Assumption 7(a), the function

$$h_2 \mapsto F_L(c_L(\rho); h_1, h_2, \rho)$$

is continuous and strictly decreasing. Therefore,

$$h_2 \mapsto \Pi_L(h_1, h_2, \rho)$$

is continuous and strictly increasing.

Next, since $c_L(\rho)$ is finite, Assumption 7(b) implies

$$\lim_{h_2 \to \infty} F_L(c_L(\rho); h_1, h_2, \rho) \;=\; 0.$$

Hence,

$$\lim_{h_2 \to \infty} \Pi_L(h_1, h_2, \rho) \;=\; 1.$$

Since $\Pi_S(h_1) < 1$, it follows that

$$\Pi_S(h_1) \;<\; \lim_{h_2 \to \infty} \Pi_L(h_1, h_2, \rho).$$

On the other hand, Assumption 7(c) gives

$$\lim_{h_2 \to -\infty} \Pi_L(h_1, h_2, \rho) \;<\; \Pi_S(h_1).$$

Define

$$D(h_2) \;=\; \Pi_L(h_1, h_2, \rho) - \Pi_S(h_1).$$

The previous arguments imply that D is continuous and strictly increasing on $\mathbb{R}$. Moreover,

$$\lim_{h_2 \to -\infty} D(h_2) \;<\; 0,$$

and

$$\lim_{h_2 \to \infty} D(h_2) \;> 0.$$

By the intermediate value theorem, there exists $\bar{h}_2 \in \mathbb{R}$ such that

$$D(\bar{h}_2) \;=\; 0.$$

Equivalently,

$$\Pi_L(h_1, \bar{h}_2, \rho) \;=\; \Pi_S(h_1).$$

Since D is strictly increasing, this zero is unique. Therefore,

$$D(h_2) < 0 \quad \text{for all } h_2 < \bar{h}_2,$$

and

$$D(h_2) > 0 \quad \text{for all } h_2 > \bar{h}_2.$$

The desired inequalities follow from the definition of D. ■

The asymptotic power of GMS inference for models applied to $\mathcal{M}_S$ and $\mathcal{M}_L$ under local alternatives can be organized into seven mutually exclusive and exhaustive cases described in Table 1. In all cases except case 3 and 7, the limits in (**??**) exist and its asymptotic behavior is as described in Table 1.[1] See Theorem **??** for the proof.

Case	Parameter configuration	Asy. power comparison	Relevant ($h_1 \geq h_2$)	winner
1	$h_1 = -\infty$ and $h_2 \geq 0$	$\pi_L > \pi_S = 0$	no	L
2	$h_1 \geq 0$, $h_2 = -\infty$, and $\eta_2 < -1$	$\pi_L = \pi_S \geq \alpha$	yes	both
3	$h_1 \geq 0$, $h_2 = -\infty$, and $\eta_2 = -1$	$\pi_L \leq \pi_S \geq \alpha$	yes	S
4	$h_1 \geq 0$, $h_2 = -\infty$, and $\eta_2 \in (-1, 0]$	$\pi_L < \pi_S \geq \alpha$	yes	S
5	$h_1 \in \mathbb{R}$ and $h_2 = \infty$	$\pi_L = 1 > \pi_S > \alpha$	no	L
6	$h_1 = \infty$ and $h_2 \in \mathbb{R} \cup \{\infty\}$	$\pi_L = \pi_S = 1$	yes	both
7	$h_1, h_2 \in \mathbb{R}$ and $\max\{h_1, h_2\} \geq 0$	$\pi_L \lesseqgtr \pi_S$	yes if $h_1 \geq h_2$	anything

Table 1: Description of the asymptotic local power behavior of the GMS inference applied to $\mathcal{M}_S$ and $\mathcal{M}_L$. The cases are mutually exclusive and exhaustive. We use π_L and π_S to informally denote the limiting rejection rate of GMS inference under $\mathcal{M}_L$ and $\mathcal{M}_S$, respectively.

We now provide a description of the cases in Table 1. In case 1, the first moment inequality is very slack (i.e. $h_1 = -\infty$), and the second one is violated (i.e. $h_2 \geq 0$). Since $\mathcal{M}_S$ only considers the first inequality, its GMS inference has no asymptotic power. On the other hand, $\mathcal{M}_L$ includes both inequalities, and so its GMS have positive asymptotic power.

In cases 2, 3, and 4, the first moment inequality is violated (i.e. $h_1 \geq 0$) and the second one is very slack (i.e. $h_2 = -\infty$). Since both $\mathcal{M}_S$ and $\mathcal{M}_L$ include the violated moment inequality, their GMS inferences have non-trivial asymptotic power. In addition, $\mathcal{M}_L$ includes the very slack moment inequality. In case 2, the second moment inequality is *extremely* slack (i.e. $\eta_2 < -1$), and the GMS method applied to $\mathcal{M}_L$ recovers this information without incurring in any loss in terms of asymptotic statistical power. On the other hand, in case 4 the second moment inequality is very slack, but not extremely so (i.e. $\eta_2 \in [-1, 0]$), and so the GMS method applied to $\mathcal{M}_L$ recovers this information at the expense of nontrivial asymptotic statistical power. Finally, case 3 is an intermediate case between cases 2 and 4, and where the GMS method can alternate its behavior between the two cases.

In case 5, the second moment inequality is very violated (i.e. $h_2 = \infty$) and the first one could be slack of violated, but not by much (i.e. $h_1 \in \mathbb{R}$). Since $\mathcal{M}_L$ includes a moment inequality that is very violated, the GMS method applied to $\mathcal{M}_L$ achieves perfect asymptotic power. On the other hand, $\mathcal{M}_S$ only includes a moment inequality that, if violated, it is not violated by much. As a result, the GMS method applied to $\mathcal{M}_S$ does not achieve perfect asymptotic power.

[1] In case 3, the limit that defines π_L need not exists, and so $\pi_L < \pi_S$ denotes that $\limsup_{n\to\infty} P(Q_{n,L}(\theta_n) > c_{n,L}(\theta_n, 1-\alpha)) \leq \pi_S$. In case 7, neither the limit that defines π_L nor the limit that defines π_S exists, and so $\pi_L \lesseqgtr \pi_S$ denotes that anything can occur depending on parameter values.

In case 6, the first moment inequality is very violated (i.e. $h_1 = \infty$) and the first one could be slack, violated, or very violated (i.e. $h_1 \in \mathbb{R} \cup \{\infty\}$). Since both $\mathcal{M}_S$ and $\mathcal{M}_L$ include the vert violated moment inequality, their GMS inferences have perfect asymptotic power. In the case of $\mathcal{M}_L$, the inclusion of an additional moment inequality does not change the asymptotic power of the GMS method due to the fact that the first moment inequality is very violated.

We conclude with case 7, which is the more complicated case in terms of possible outcomes. In this case, at least one of the moment inequalities is violated ($\max\{h_1, h_2\} \geq 0$), but neither is either very violated or very slack (i.e. $h_1, h_2 \in \mathbb{R}$). As a result, the relative asymptotic power of the GMS method depends on the limiting parameters of the data: h_1, h_2, and ρ. Depending on these parameters, we reproduce the behavior that is similar to case 2 (i.e. $\pi_L < \pi_S$), case 4 (for which $\pi_L > \pi_S$), and case 5 (i.e. $\pi_L = \pi_S$). It is interesting to note that it is possible for GMS inference applied to $\mathcal{M}_S$ to be strictly more powerful than GMS inference applied to $\mathcal{M}_L$ (i.e. yellow region) even in cases in which the moment inequality ignored by $\mathcal{M}_S$ is violated (i.e. $h_2 > 0$). For example, see $(h_1, h_2) = (3, 0.5)$.

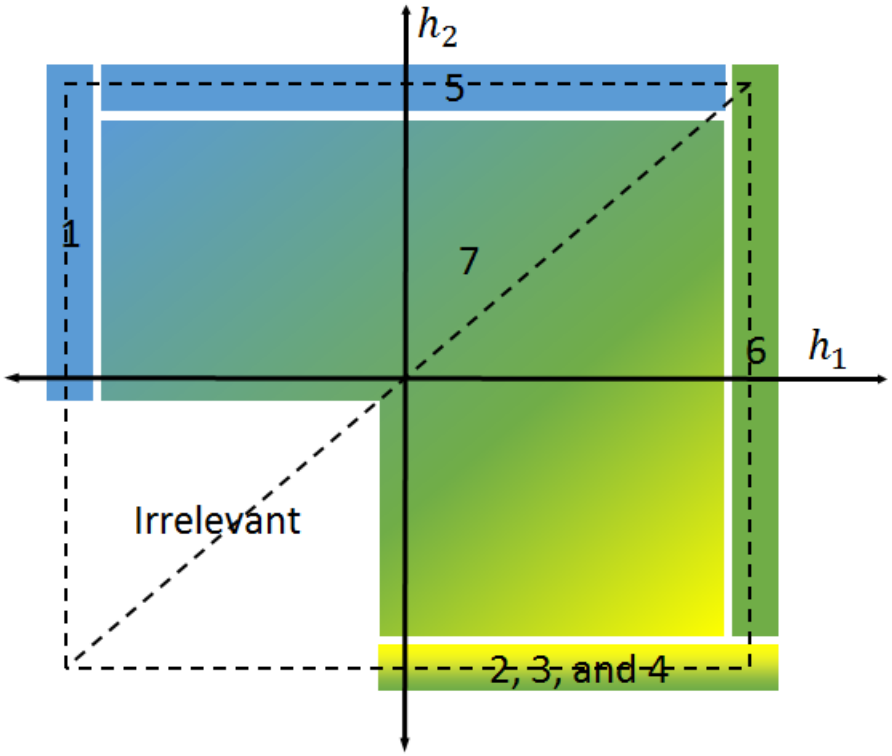


Figure 1: Graphical description of the possible cases for local power analysis in Theorem **??**. The blue region represents a situation in which the asymptotic power difference between $\mathcal{M}_L$ and $\mathcal{M}_S$ is positive, the yellow region represents a situation in which the asymptotic power difference between $\mathcal{M}_L$ and $\mathcal{M}_S$ is negative, and green region represents a situation in which $\mathcal{M}_L$ and $\mathcal{M}_S$ have the same asymptotic power. Note that Case 3 and Case 4 are identical in terms of (h_1, h_2).

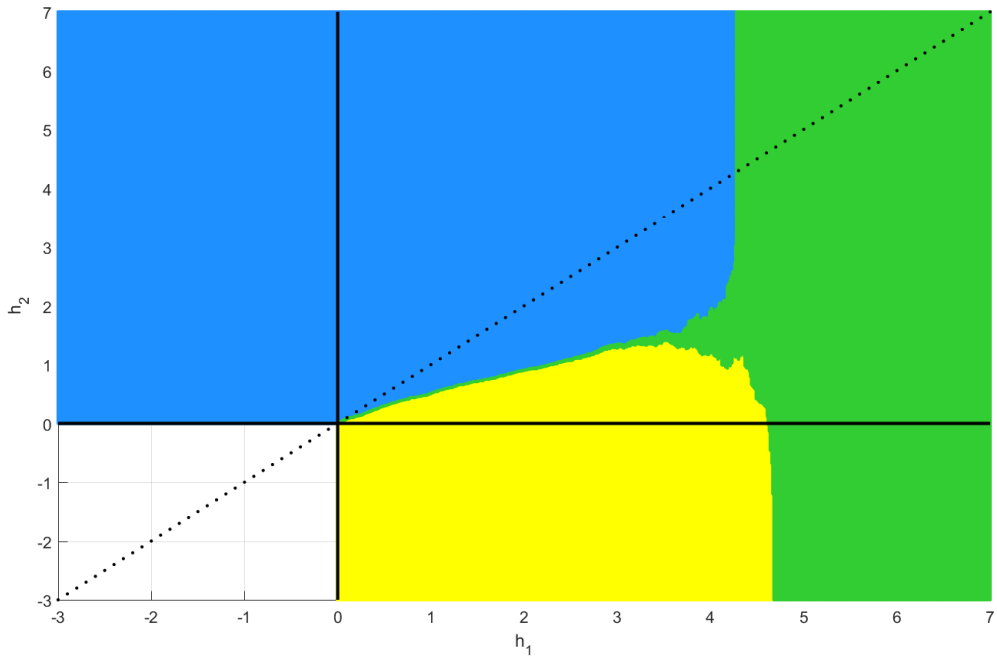


Figure 2: Graphical description of the relative asymptotic power of the GMS method applied to $\mathcal{M}_L$ and $\mathcal{M}_S$ when $\rho = 0$ and $\alpha = 0.1$. The blue region represents a situation in which the asymptotic power advantage of $\mathcal{M}_L$ over $\mathcal{M}_S$ exceeds 0.1%, the yellow region represents a situation in which the asymptotic power advantage of $\mathcal{M}_S$ over $\mathcal{M}_L$ exceeds 0.1%, and green region represents a situation in which the asymptotic power difference between $\mathcal{M}_L$ and $\mathcal{M}_S$ is less than 0.1%.

3 Monte Carlo simulations

This section presents Monte Carlo simulations based on the econometric model described in Section 2. We simulate independent samples with $n = 1,000$. We consider the econometric model in (2.1) with $\theta = 0$ and P in (**??**) with $\sigma_{1,P} = \sigma_{2,P} = 1$, $\rho_P = 0$, $\mu_{1,P} = -1/\sqrt{1,000}$, and $\mu_{2,P} = -h_2/\sqrt{1,000}$ for $h_2 \in \{-1, -0.5, 0, 0.5, 1, 1.5, 2\}$. As a consequence, $h_1 = 1$ and $h_2 \in \{-1, -0.5, 0, 0.5, 1, 1.5, 2\}$, i.e., the first moment inequality is not satisfied and the second moment inequality is either slack (when $h_2 = \{-1, -0.5\}$), satisfied with equality (when $h_2 = 0$), or violated (when $h_2 \in \{0.5, 1, 1.5, 2\}$). This configuration of parameters corresponds to Case 7 in Table 1 and Figure 2.

According to the asymptotic approximations described in Figure 2, GMS inference based on $\mathcal{M}_S$ should be relatively more powerful than GMS inference based on $\mathcal{M}_L$ for negative and positive but low values of h_2, but this should exactly reverse for large positive values of h_2. Table 2 confirms that these predictions hold in the Monte Carlo simulations. Note that $\mathcal{M}_S$ only considers the first moment inequality and so the corresponding GMS inference does not change with h_2. On the other hand, $\mathcal{M}_L$ incorporates information from both moment inequalities. The empirical rejection rate

of the GMS inference based on $\mathcal{M}_L$ increases monotonically with the amount of (local) violation in the second moment condition, given by h_2. When the second moment inequality is mildly violated, the average increase in the GMS test function does not compensate for the effect of a larger GMS critical value, resulting in a relatively lower power of the GMS inference based on $\mathcal{M}_L$. When the second moment inequality is sufficiently violated, the average increase in the GMS test function can more than compensate for the effect of a larger GMS critical value, resulting in a relatively higher power of the GMS inference based on $\mathcal{M}_L$.

h_2	Avg. $Q_{n,S}(\theta)$	Avg. $c_{n,S}(\theta)$	Rej. rate for $\mathcal{M}_S$ (in %)	Avg. $Q_{n,L}(\theta)$	Avg. $c_{n,L}(\theta)$	Rej. rate for $\mathcal{M}_L$ (in %)
-1	1.84	1.64	37.90	1.91	2.87	24.50
-0.5	1.84	1.64	37.90	2.03	2.93	25.10
0	1.84	1.64	37.90	2.31	2.94	29.05
0.5	1.84	1.64	37.90	2.85	2.95	37.10
1	1.84	1.64	37.90	3.74	2.95	49.60
1.5	1.84	1.64	37.90	5.04	2.95	65.10
2	1.84	1.64	37.90	6.81	2.95	78.50

Table 2: Empirical results from implementing GMS inference for models $\mathcal{M}_L$ and $\mathcal{M}_S$. We report results from $S = 2,000$ Monte Carlo simulations, each with sample size $n = 1,000$. The parameters are $\theta = 0$ and P in (**??**) with $\sigma_{1,P} = \sigma_{2,P} = 1$, $\rho_P = 0$, $\mu_{1,P} = -1/\sqrt{1,000}$, and $\mu_{2,P} = -h_2/\sqrt{1,000}$ for $h_2 \in \{-1, -0.5, 0, 0.5, 1, 1.5, 2\}$.

4 Recommendations on the choice of the moment (in)equalities (OPTIONAL)

SO FAR, THIS PAPER SHOWS THAT adding or removing moment conditions has an ambiguous effect on the asymptotic power of the inference in a partially identified moment (in)equality model. IT ALL DEPENDS ON THE LOCAL ALTERNATIVE UNDER CONSIDERATION. TO ME, THIS RESULT SOUNDS TOO NEGATIVE AND VAGUE. TO CLOSE THE PAPER, I WOULD PREFER TO GIVE A MORE POSITIVE AND CONSTRUCTIVE MESSAGE.

MY IDEA: RATHER THAT LOOK AT SPECIFIC SEQUENCES OF LOCAL ALTERNATIVES, CONSIDER AN AVERAGE POWER OF OVER LOCAL ALTERNATIVES. TO SELL THIS IDEA, WE NEED TWO THINGS:

- THE AVERAGE SHOULD BE BASED ON A REASONABLE AND FAIR DISTRIBUTION.
- THE ANALYSIS SHOULD PROVIDE A CONCRETE RECOMMENDATION.

What is a reasonable distribution over local alternatives? Consider the model in Section 2. The distribution should provide weights over over possible values of h_1 and h_2 which, in turn, determine the average values of π_L and π_S. Being "fair" means that h_1 and h_2 should be treated symmetrically.

If we choose a real-valued distribution of h_1 and h_2, then we are automatically in Case 7 in terms of Table 1.

CONJECTURE: I conjecture that this will favor the larger model. This occurs in the particular example in Section 2 regardless of ρ. I think this can be shown in general. I have an idea for a proof strategy, but have not tried it yet.

5 Statements to prove

At the risk of slightly deviating from the notation in the rest of these notes, let's define the following objects:

- $\phi_n^{\mathrm{s}}(X)$ is the "short" test that only use the minimal set of inequalities that characterize the identified set. In practice, we will focus on the test that uses a max-type test statistic and two alternative critical values (and self-normalized and one that uses moment selection).
- $\phi_n^{\mathrm{l}}(X)$ is the "long" test that uses ALL the inequalities that characterize the identified set.
- $\phi_n^{\mathrm{m}}(X, \zeta_n)$ is the mixing test that is defined as follows. Let $\zeta_n(X)$ be a function that maps the support of the data to the interval $[0, 1]$. Note that ζ_n can take values strictly above 0 and below 1. Define

$$\phi_n^{\mathrm{m}}(X, \zeta_n) = \zeta_n(X)\phi_n^{\mathrm{s}}(X) + (1 - \zeta_n(X))\phi_n^{\mathrm{l}}(X). \tag{5.1}$$

  When the mixing function $\zeta_n(X)$ is strictly in $(0, 1)$, the mixing test is one that equals the short test with probability $\zeta_n(X)$ and that equals the long test with probability $1 - \zeta_n(X)$.

We wish to show the following two results.

Theorem 5.1. Under certain assumptions, there exist $\zeta_n^{\mathrm{s}}(X)$ and $\zeta_n^{\mathrm{l}}(X)$ such that

$$E_P[\phi_n^{\mathrm{m}}(X, \zeta_n^{\mathrm{s}})] \geq E_P[\phi_n^{\mathrm{s}}(X)] \quad \forall P \in \mathbf{P}_1 \ , \tag{5.2}$$

with strict inequality for some "subset?" of $\mathbf{P}_1$, and

$$E_P[\phi_n^{\mathrm{m}}(X, \zeta_n^{\mathrm{l}})] \geq E_P[\phi_n^{\mathrm{l}}(X)] \quad \forall P \in \mathbf{P}_1 \ . \tag{5.3}$$

with strict inequality for some "subset?" of $\mathbf{P}_1$.

Note that an important feature of the above result is that the "optimal" mixing rules ζ_n^{s} and ζ_n^{l} may be arbitrarily different.

Theorem 5.2. Under certain assumptions, $\nexists \zeta_n^*$ such that

$$E_P[\phi_n^{\mathrm{m}}(X, \zeta_n^*)] \geq \max\{E_P[\phi_n^{\mathrm{s}}(X)], E_P[\phi_n^{\mathrm{l}}(X)]\} \quad \forall P \in \mathbf{P}_1 \ . \tag{5.4}$$

The idea is that there is no "optimal" mixing rule ζ_n^* that is capable of beating the infeasible best test (defined as the maximum of the two known tests).

6 Conclusions

HERE ARE MY TAKEAWAYS FROM THIS PAPER SO FAR:

- Adding/removing a moment (in)equality has an ambiguous effect on the local asymptotic power of GMS inference. This can occur even if the moment (in)equality that is added/removed is (locally) violated.
- This finding is qualitatively different from Moon and Schorfheide (2009). The main reason behind the difference is that Moon and Schorfheide (2009) study a point identified moment (in)equality model (rather than partially identified) and conducts inference using empirical likelihood (rather than GMS).
- When does adding a violated moment (in)equality model increase the asymptotic power of inference? This depends on the Studentized magnitude of $E_P[m_j(W,\theta)]/\sqrt{Var_P[m_j(W,\theta)]}$. The literature on "core determining sets" focuses on $E_P[m_j(W,\theta)]$ but largely ignores the effect of $Var_P[m_j(W,\theta)]$.

A Appendix

Lemma A.1. For a sequence of arbitrary distributions $\{P_n : n \in \mathbb{N}\}$, assume that the random variables T_n and c_n satisfy the following conditions:

(a) $T_n \xrightarrow{d} T$ along $\{P_n : n \in \mathbb{N}\}$, i.e., for every x s.t. $P(T \leq x)$ is continuous, $P_n(T_n \leq x) \to P(T \leq x)$,

(b) $P(T \leq x)$ is continuous at c,

(c) $c_n \xrightarrow{p} c$ along $\{P_n : n \in \mathbb{N}\}$, i.e., $c_n = c + o_{P_n}(1)$.

Then, $P_n(T_n > c_n) \to P(T > c)$ as $n \to \infty$.

Proof. Fix $\varepsilon > 0$ such that $c + \varepsilon$ and $c - \varepsilon$ are continuity points of $P(T > h)$. Note that

$$\begin{aligned} P_n(T_n > c_n) &= P_n(T_n > c_n \cap |c_n - c| \leq \varepsilon) + P_n(T_n > c_n \cap |c_n - c| > \varepsilon) \\ &\leq P_n(T_n > c - \varepsilon) + P_n(|c_n - c| > \varepsilon). \end{aligned} \tag{A.1}$$

By a similar argument, we get

$$P_n(T_n > c_n) \geq P_n(T_n > c + \varepsilon) + P_n(|c_n - c| > \varepsilon). \tag{A.2}$$

Then, consider the following argument.

$$\begin{aligned} P(T > c + \varepsilon) &\overset{(1)}{=} \liminf_{n\to\infty} P_n(T_n > c + \varepsilon) + \liminf_{n\to\infty} P_n(|c_n - c| > \varepsilon) \\ &\overset{(2)}{\leq} \liminf_{n\to\infty} P_n(T_n > c_n) \\ &\leq \limsup_{n\to\infty} P_n(T_n > c_n) \\ &\overset{(3)}{\leq} \limsup_{n\to\infty} P_n(T_n > c - \varepsilon) + \limsup_{n\to\infty} P_n(|c_n - c| > \varepsilon) \\ &\overset{(4)}{=} P(T > c - \varepsilon), \end{aligned} \tag{A.3}$$

where (1) holds by $c_n \xrightarrow{p} c$, $T_n \xrightarrow{d} T$, and that $c + \varepsilon$ is a continuity point of $P(T > h)$, (2) by taking limit inferior as $n \to \infty$ on (A.2), (3) by taking limit superior as $n \to \infty$ in (A.1), and (4) by $c_n \xrightarrow{p} c$, $T_n \xrightarrow{d} T$, and that $c - \varepsilon$ is a continuity point of $P(T > h)$.

Note that (A.3) holds for all $\varepsilon > 0$ such that $c + \varepsilon$ and $c - \varepsilon$ are continuity points of $P(T > h)$. By elementary properties of any CDF, we can find a sequence $\{\varepsilon_m : m \geq 1\}$ such that $\varepsilon_m \downarrow 0$ and $c + \varepsilon_m$ and $c - \varepsilon_m$ are continuity points of $P(T > h)$ for all $m \in \mathbb{N}$. Then, (A.3) implies that for all $m \in \mathbb{N}$,

$$P(T > c + \varepsilon_m) \leq \liminf_{n\to\infty} P_n(T_n > c_n) \leq \limsup_{n\to\infty} P_n(T_n > c_n) \leq P(T > c - \varepsilon_m). \tag{A.4}$$

By taking limit superior and limit inferior as $m \to \infty$ on (A.4) and using that the CDF of T is continuous at c, the desired result follows. ■

Lemma A.2. Consider GMS inference applied to (2.1), where S satisfies Assumption 1. Assume that the sequence $\{(P_n, \theta_n) \in \mathcal{P} \times \Theta\}_{n \geq 1}$ with $\rho_{P_n} \to \rho$, $\sigma_{j,P_n} \to \sigma_j > 0$ for $j = 1, 2$, and $\sqrt{n}(\theta_n - \mu_{j,n})/\sigma_{j,n} \to h_j \in \mathbb{R}_{[+\infty]}$ for $j = 1, 2$. Then,

1. $S(\sqrt{n}(\theta_n - \bar{W}_1), \hat{\sigma}_1^2) \xrightarrow{d} S(\xi_1 + h_1, 1)$ along $\{P_n\}$, where $\xi_1 \sim N(0, 1)$.
2. $S((\sqrt{n}(\theta_n - \bar{W}_1), \sqrt{n}(\theta_n - \bar{W}_2))', [\hat{\sigma}_1^2, \hat{\sigma}_1\hat{\sigma}_2\hat{\rho}; \hat{\sigma}_1\hat{\sigma}_2\hat{\rho}, \hat{\sigma}_2^2]) \xrightarrow{d} S((\xi_1 + h_1, \xi_2 + h_2)', [1, \rho; \rho, 1])$ along $\{P_n\}$, where $(\xi_1, \xi_2) \sim N(\mathbf{0}_2, [1, \rho; \rho, 1])$.

Proof. For part 1, consider the following derivation.

$$
\begin{aligned}
& S(\sqrt{n}(\theta_n - \bar{W}_1), \hat{\sigma}_1^2) \\
& \overset{(1)}{=} \left\{ \begin{array}{c} S([-\sqrt{n}(\bar{W}_1 - \mu_{1,P_n})/\sigma_{1,P_n} + \sqrt{n}(\theta_n - \mu_{1,P_n})/\sigma_{1,P_n}](\sigma_{1,P_n}/\hat{\sigma}_1), 1) I[\hat{\sigma}_1 > 0] \\ + S(\sqrt{n}(\theta_n - \bar{W}_1), \hat{\sigma}_1^2) I[\hat{\sigma}_1 = 0] \end{array} \right\} \\
& \overset{(2)}{\xrightarrow{d}} S(\xi_1 + h_1, 1),
\end{aligned}
$$

where (1) holds by Assumption 1(b), and (2) by Assumption 1(c), the CMT, and the triangular-array CLT and LLN which imply that $\sqrt{n}(\bar{W}_1 - \mu_{1,P_n})/\sigma_{1,P_n} \xrightarrow{d} \xi_1$ and $\hat{\sigma}_1/\sigma_{1,P_n} = 1 + o_{P_n}(1)$ along $\{P_n \in \mathcal{P}\}_{n \geq 1}$.

For part 2, consider the following derivation.

$$
\begin{aligned}
& S(\left(\sqrt{n}(\theta_n - \bar{W}_1), \sqrt{n}(\theta_n - \bar{W}_2)\right)', [\hat{\sigma}_1^2, \hat{\rho}\hat{\sigma}_1\hat{\sigma}_2; \hat{\rho}\hat{\sigma}_1\hat{\sigma}_2, \hat{\sigma}_2^2]) \\
& \overset{(1)}{=} \left\{ \begin{array}{c} S\left(\left[\begin{array}{c} \left(\sqrt{n}(\mu_{1,P_n} - \bar{W}_1)/\sigma_{1,P_n} + \sqrt{n}(\theta_n - \mu_{1,P_n})/\sigma_{1,P_n}\right)(\sigma_{1,P_n}/\hat{\sigma}_1) \\ \left(\sqrt{n}(\mu_{2,P_n} - \bar{W}_2)/\sigma_{2,P_n} + \sqrt{n}(\theta_n - \mu_{2,P_n})/\sigma_{2,P_n}\right)(\sigma_{2,P_n}/\hat{\sigma}_2) \end{array} \right], \left[\begin{array}{cc} 1 & \hat{\rho} \\ \hat{\rho} & 1 \end{array} \right] \right) \\ \times I[\hat{\sigma}_1, \hat{\sigma}_2 > 0] \\ + S(\left(\sqrt{n}(\theta_n - \bar{W}_1), \sqrt{n}(\theta_n - \bar{W}_2)\right)', [\hat{\sigma}_1^2, \hat{\rho}\hat{\sigma}_1\hat{\sigma}_2; \hat{\rho}\hat{\sigma}_1\hat{\sigma}_2, \hat{\sigma}_2^2])(1 - I[\hat{\sigma}_1, \hat{\sigma}_2 > 0]) \end{array} \right\} \\
& \overset{(2)}{\xrightarrow{d}} S((\xi_1 + h_1, \xi_2 + h_2)', [1, \rho; \rho, 1]),
\end{aligned}
$$

where (1) holds by Assumption 1(b), and (2) by Assumption 1(c), the CMT, and the triangular-array CLT and LLN which imply that $\left(\sqrt{n}(\mu_{1,P_n} - \bar{W}_1)/\sigma_{1,P_n}, \sqrt{n}(\mu_{2,P_n} - \bar{W}_2)/\sigma_{2,P_n}\right) \xrightarrow{d} (\xi_1, \xi_2)$, $\hat{\sigma}_j/\sigma_{j,P_n} = 1 + o_{P_n}(1)$ for $j = 1, 2$, $\hat{\rho} = \rho + o_{P_n}(1)$ along $\{P_n \in \mathcal{P}\}_{n \geq 1}$. ■

Lemma A.3. Consider GMS inference applied to (2.1), where S satisfies Assumption 1-2 and φ satisfies Assumption 3. Assume that the sequence $\{(P_n, \theta_n) \in \mathcal{P} \times \Theta\}_{n \geq 1}$ with $\rho_{P_n} \to \rho$, $\sigma_{j,P_n} \to \sigma_j > 0$ for $j = 1, 2$, and $\kappa_n^{-1}\sqrt{n}(\theta_n - \mu_{j,n})/\sigma_{j,n} \to \eta_j \in C$ for $j = 1, 2$, where C and κ_n are as specified in Assumption 3. Furthermore, consider the CDFs:

$$
\begin{aligned}
F_S(x) &= P\left(S\left(\xi_1 + \varphi_1\left(-\eta_1, 1\right), 1\right) \leq x\right) \\
F_L(x) &= P\left(S\left(\xi_1 + \varphi_1((-\eta_1, -\eta_2), \Omega), \xi_2 + \varphi_2((-\eta_2, -\eta_1), \Omega), \Omega\right) \leq x\right),
\end{aligned}
$$

with $(\xi_1, \xi_2) \sim N(\mathbf{0}_2, \Omega)$ and $\Omega = [1, \rho; \rho, 1]$ are strictly increasing at their $(1 - \alpha)$-quantiles, denoted by $c_S(\eta_1)$ and $c_L(\eta_1, \eta_2, \rho)$, respectively. Then,

(a) $\hat{c}_{n,S}(\theta_n) \xrightarrow{p} c_S(\eta_1)$ along $\{P_n : n \in \mathbb{N}\}$.

(b) $\hat{c}_{n,L}(\theta_n) \xrightarrow{p} c_L(\eta_1, \eta_2, \rho)$ along $\{P_n : n \in \mathbb{N}\}$.

Proof. We simultaneously prove the result for $p = 1, 2$. That is, we show $\hat{c}_n(\theta_n) \xrightarrow{p} c(\eta, \Omega)$ along $\{P_n : n \in \mathbb{N}\}$, where $\hat{c}_n(\theta_n)$ is the conditional $(1-\alpha)$-quantile of $Q_n^*(\theta_n) = S(v_n^* + \varphi(\hat{\xi}(\theta_n), \hat{\Omega}), \hat{\Omega})$, $\varphi(\xi, \Omega) = \{\varphi_a(\xi, \Omega) : a = 1, \ldots, p\}$, $\hat{\xi}(\theta) = \{\hat{\xi}_a(\theta) : a = 1, \ldots, p\}$, $\hat{\xi}_a(\theta) = \kappa_n^{-1}\sqrt{n}(\bar{W}_a - \theta)/\hat{\sigma}_a I\{\hat{\sigma}_a > 0\}$ for $a = 1, \ldots, p$, $v_n^* = \{v_{n,a}^* : a = 1, \ldots, p\}$ with $v_{n,a}^* = n^{-1/2}\sum_{i=1}^n \zeta_i(W_{i,a} - \bar{W}_a)/\hat{\sigma}_a I\{\hat{\sigma}_a > 0\}$ for $a = 1, \ldots, p$, and $c(\eta, \Omega)$ is the $(1-\alpha)$-quantile of $\Pi(x) = P(S(\xi + \varphi(-\eta, \Omega), \Omega) \le x)$, where $\xi \sim N(\mathbf{0}_p, \Omega)$, $\eta = \{\eta_a : 1, \ldots, p\}$, and $\varphi(-\eta, \Omega) = \{\varphi_a(-\eta, \Omega) : 1, \ldots, p\}$. Part (a) corresponds to $p = 1$, while part (b) corresponds to $p = 2$.

Fix $\delta \in (0, 1)$ arbitrarily. Let

$$E_n(\delta) \equiv \left\{ \begin{array}{c} \{|\kappa_n^{-1}\sqrt{n}(\bar{W}_a - \mu_{a,P_n})/\sigma_a| \le \delta : a = 1, \ldots, p\} \cap \\ \{\hat{\sigma}_a > 0 : a = 1, \ldots, p\} \cap \{|\hat{\sigma}_a/\sigma_a - 1| \le \delta : a = 1, \ldots, p\} \\ \cap \{|\hat{\Omega}_{[a,b]} - \Omega_{[a,b]}| \le \delta : a, b = 1, \ldots, p\} \end{array} \right\}.$$

By the triangular array CLT and LLN, we have that, along any sequence $\{P_n \in \mathcal{P}\}_{n\ge1}$, $\sqrt{n}(\bar{W}_a - \mu_{a,n})/\sigma_{a,P_n} \xrightarrow{d} N(0,1)$, $\hat{\sigma}_a/\sigma_{a,P_n} = 1 + o_{P_n}(1)$, $|\hat{\Omega}_{[a,b]} - \Omega_{P_n,[a,b]}| = o_{P_n}(1)$ for $a, b = 1, \ldots, p$. Along $\{P_n \in \mathcal{P}\}_{n\ge1}$, $\sigma_{a,P_n} \to \sigma_a > 0$ and $|\Omega_{P_n,[a,b]} - \Omega_{[a,b]}| \to 0$ for $a, b = 1, \ldots, p$. By combining these and $\kappa_n \to \infty$ (by Assumption 3(d)), we get

$$P_n(E_n(\delta)) \to 1. \tag{A.5}$$

Let $N \in \mathbb{N}$ be such that $\{|\kappa_n^{-1}\sqrt{n}(\theta_n - \mu_{a,P_n})/\sigma_a - \eta_a| \le \delta : a = 1, \ldots, p\}$. For any $n \ge N$,

$$\begin{aligned}
&P_n(Q_n^*(\theta_n) \le h \mid \{W_i\}_{i=1}^n) \\
&= P_n(S(v_n^* + \varphi(\hat{\xi}, \hat{\Omega}), \hat{\Omega}) \le h \mid \{W_i\}_{i=1}^n) \\
&\overset{(1)}{=} P_n\left(S\left(v_n^* + \varphi\left(\left\{ \begin{array}{c} (\kappa_n^{-1}\sqrt{n}(\bar{W}_j - \mu_{j,P_n})/\sigma_j)(\sigma_j/\hat{\sigma}_j) + \\ (\kappa_n^{-1}\sqrt{n}(\mu_{j,P_n} - \theta_n)/\sigma_j)(\sigma_j/\hat{\sigma}_j - 1) \\ + (\kappa_n^{-1}\sqrt{n}(\mu_{j,P_n} - \theta_n)/\sigma_j) \end{array} \right\}_{j=1}^p, \hat{\Omega} \right), \hat{\Omega} \right) \le h \middle| \{W_i\}_{i=1}^n \right) I\{E_n(\delta)\} \\
&\qquad + R_{n,1} \\
&\overset{(2)}{\le} P_n\Big(\inf_{\Delta_2 \in \Lambda(\delta)} S\big(v_n^* + \sup_{\pi \in B(\delta), \Delta_1 \in \Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega + \Delta_1), \Omega + \Delta_2\big) \le h \Big| \{W_i\}_{i=1}^n \Big) I\{E_n(\delta)\} \\
&\qquad + R_{n,1} \\
&\overset{(3)}{=} P_n\Big(\inf_{\Delta_2 \in \Lambda(\delta)} S\big(v_n^* + \sup_{\pi \in B(\delta), \Delta_1 \in \Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega + \Delta_1), \Omega + \Delta_2\big) \le h \Big| \{W_i\}_{i=1}^n \Big) \\
&\qquad + R_{n,1} + R_{n,2},
\end{aligned} \tag{A.6}$$

where (1) and (3) hold by defining

$$\begin{aligned}
R_{n,1} &= P_n(S(v_n^* + \varphi(\hat{\xi}, \hat{\Omega}), \hat{\Omega}) \le h \mid \{W_i\}_{i=1}^n) I\left\{E_n\left(\delta\right)^c\right\} \\
R_{n,2} &= P_n\Big(\inf_{\Delta_2 \in \Lambda(\delta)} S\big(v_n^* + \sup_{\pi \in B(\delta), \Delta_1 \in \Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega + \Delta_1), \Omega + \Delta_2\big) \le h \Big| \{W_i\}_{i=1}^n \Big) I\left\{E_n\left(\delta\right)^c\right\},
\end{aligned}$$

and (2) by Assumption 1(a) and defining:

$$B(\delta) = \{x \in \mathbb{R}^p : x_j = \phi_1(1 + \phi_2) + (\phi_3 - \eta_j)\phi_4 + \phi_5 \text{ for some } \{\phi_j \in \mathbb{R} : |\phi_j| \le \delta,\ j = 1, \ldots, 5\}\},$$

$$\Lambda(\delta) = \{x \in \mathbb{R}^{k\times k} : \|x_j\| < \delta\}.$$

Note that A.5 implies that $R_{n,1} + R_{n,2} = o_{P_n}(1)$. By a similar argument, for any $n \geq N$,

$$P_n(Q^*_{n,S}(\theta_n) \leq h \mid \{W_i\}_{i=1}^n) \geq$$
$$P_n\Big(\sup_{\Delta_2\in\Lambda(\delta)} S\big(v^*_n + \inf_{\pi\in B(\delta),\Delta_1\in\Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega+\Delta_1), \Omega+\Delta_2\big) \leq h \Big| \{W_i\}_{i=1}^n\Big) + R_{n,1} + R_{n,3}, \quad \text{(A.7)}$$

with $R_{n,1} + R_{n,3} = o_{P_n}(1)$, where we define

$$R_{n,3} = P_n\Big(\sup_{\Delta_2\in\Lambda(\delta)} S\big(v^*_n + \inf_{\pi\in B(\delta),\Delta_1\in\Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega+\Delta_1), \Omega+\Delta_2\big) \leq h \Big| \{W_i\}_{i=1}^n\Big) I\{E_n(\delta)^c\}.$$

For arbitrary $h \in \mathbb{R}$, we now show that:

$$\begin{aligned} &P_n\Big(\inf_{\Delta_2\in\Lambda(\delta)} S\big(v^*_n + \sup_{\pi\in B(\delta),\Delta_1\in\Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega+\Delta_1), \Omega+\Delta_2\big) \leq h \Big| \{W_i\}_{i=1}^n\Big) \\ &= P\Big(\inf_{\Delta_2\in\Lambda(\delta)} S\big(\xi + \sup_{\pi\in B(\delta),\Delta_1\in\Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega+\Delta_1), \Omega+\Delta_2\big) \leq h\Big) + R_{n,4}, \end{aligned} \quad \text{(A.8)}$$

with $R_{n,4} = o_{P_n}(1)$, and

$$\begin{aligned} &P_n\Big(\sup_{\Delta_2\in\Lambda(\delta)} S\big(v^*_n + \inf_{\pi\in B(\delta),\Delta_1\in\Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega+\Delta_1), \Omega+\Delta_2\big) \leq h \Big| \{W_i\}_{i=1}^n\Big) \\ &= P\Big(\sup_{\Delta_2\in\Lambda(\delta)} S\big(\xi + \inf_{\pi\in B(\delta),\Delta_1\in\Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega+\Delta_1), \Omega+\Delta_2\big) \leq h\Big) + R_{n,5}, \end{aligned} \quad \text{(A.9)}$$

with $R_{n,5} = o_{P_n}(1)$. To this end, define the functions $F_1 : \mathbb{R}^p \to \mathbb{R}$ and $F_2 : \mathbb{R}^p \to \mathbb{R}$ as follows:

$$\begin{aligned} F_1(y) &= \inf_{\Delta_2\in\Lambda(\delta)} S\big(y + \sup_{\pi\in B(\delta),\Delta_1\in\Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega+\Delta_1), \Omega+\Delta_2\big), \\ F_2(y) &= \sup_{\Delta_2\in\Lambda(\delta)} S\big(y + \inf_{\pi\in B(\delta),\Delta_1\in\Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega+\Delta_1), \Omega+\Delta_2\big). \end{aligned}$$

Under Assumption 1(a), F_1 and F_2 are continuous. Then, (A.8) and (A.9) follow from the CMT, provided that

$$\{v^*_n | \{W_i\}_{i=1}^n\} \xrightarrow{d} \xi \sim N(\mathbf{0}_p, \Omega) \text{ w.p.a.1.} \quad \text{(A.10)}$$

To conclude the proof of (A.8) and (A.9), we now establish (A.10). Conditional on $\{W_i\}_{i=1}^n$, $\{\{\xi_i(W_{a,i} - \bar{W}_a) : a = 1,\ldots,p\} : 1,\ldots,n\}$ is an independent sample with $\{\xi_i(W_{i,a} - \bar{W}_a) : a = 1,\ldots,p\} \sim N(\mathbf{0}_p, M_i)$ and $M_{i,[a,b]} = (W_{a,i} - \bar{W}_a)(W_{b,i} - \bar{W}_b)$ for all $a, b = 1,\ldots,p$. This implies that $\{\tilde{v}^*_n \mid \{W_i\}_{i=1}^n\} \sim N(\mathbf{0}_p, \hat{\Sigma})$, where $\tilde{v}^*_n = \{n^{-1/2}\sum_{i=1}^n (W_{a,i} - \bar{W}_a)\zeta_i : a = 1,\ldots,p\}$. Next, let $E_n = \{\hat{\sigma}_a > 0 : a = 1,\ldots,p\}$, and note that for any $A \subseteq \mathbb{R}^p$,

$$\begin{aligned} P(v^*_n \in A \mid \{W_i\}_{i=1}^n) &= P(v^*_n \in A \mid \{W_i\}_{i=1}^n) I[E_n] + P(v^*_n \in A \mid \{W_i\}_{i=1}^n) I[E_n^c] \\ &\overset{(1)}{=} P(N(\mathbf{0}_p, \hat{\Omega}) \in A \mid \{W_i\}_{i=1}^n) I[E_n] + P(v^*_n \in A \mid \{W_i\}_{i=1}^n) I[E_n^c] \\ &\overset{(2)}{=} P(N(\mathbf{0}_p, \Omega) \in A) + R_{n,6} + R_{n,7}, \end{aligned} \quad \text{(A.11)}$$

where (1) holds by $\{\tilde{v}_n^* \mid \{W_i\}_{i=1}^n\} \sim N(\mathbf{0}_p, \hat{\Sigma})$, and the fact that, under E_n, $v_n^* = \hat{D}^{-1}\tilde{v}_n^*$ and $\hat{\Omega} = \hat{D}^{-1}\hat{\Sigma}\hat{D}^{-1}$ with $\hat{D} = \text{diag}\{\hat{\sigma}_1, \ldots, \hat{\sigma}_p\}$, and (2) by defining

$$R_{n,6} = \big(P(N(\mathbf{0}_p, \hat{\Omega}) \in A \mid \{W_i\}_{i=1}^n) - P(N(\mathbf{0}_p, \Omega) \in A)\big) I[E_n],$$
$$R_{n,7} = \big(P(v_n^* \in A \mid \{W_i\}_{i=1}^n) - P(N(\mathbf{0}_p, \Omega) \in A)\big) I[E_n^c].$$

Since $|\hat{\Omega}_{[a,b]} - \Omega_{P_n,[a,b]}| = o_{P_n}(1)$ and $|\Omega_{P_n,[a,b]} - \Omega_{[a,b]}| \to 0$ for $a, b = 1, \ldots, p$, and by elementary properties of the normal distribution, $R_{n,6} = o_{P_n}(1)$. Since $\hat{\sigma}_a - \sigma_{a,P_n} = o_{P_n}(1)$ and $\sigma_{a,P_n} \to \sigma_a > 0$ for $a = 1, \ldots, p$, $R_{n,7} = o_{P_n}(1)$. By $R_{n,6} + R_{n,7} = o_{P_n}(1)$ and (A.11), (A.10) follows, as desired.

Next, fix $\varepsilon > 0$ arbitrarily. Since $\Pi(h)$ is a CDF that is strictly increasing at $c(\eta, \Omega)$, we have that

$$\Pi(c(\eta, \Omega) + \varepsilon) > 1 - \alpha + \lambda \text{ and } \Pi(c(\eta, \Omega) - \varepsilon) < 1 - \alpha - \lambda. \tag{A.12}$$

By (A.12), $\eta_j \in C$ for $j = 1, \ldots, p$, and Assumptions 1(a) and 3(a), $\exists \delta > 0$ small enough s.t.

$$P\Big(\inf_{\Delta_2 \in \Lambda(\delta)} S\big(\xi + \sup_{\pi \in B(\delta), \Delta_1 \in \Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega + \Delta_1), \Omega + \Delta_2\big) \le c(\eta, \Omega) + \varepsilon\Big) > 1 - \alpha + \lambda, \tag{A.13}$$
$$P\Big(\sup_{\Delta_2 \in \Lambda(\delta)} S\big(\xi + \inf_{\pi \in B(\delta), \Delta_1 \in \Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega + \Delta_1), \Omega + \Delta_2\big) \le c(\eta, \Omega) - \varepsilon\Big) < 1 - \alpha - \lambda. \tag{A.14}$$

Next, consider the following argument.

$$\begin{aligned}
&\{\hat{c}_n(\theta_n) > c(\eta, \Omega) + \varepsilon\} \\
&\subseteq \{P_n(Q_n^*(\theta_n) \le c(\eta, \Omega) + \varepsilon | \{W_i\}_{i=1}^n) < 1 - \alpha\} \\
&\overset{(1)}{\subseteq} \left\{ \begin{array}{c} P_n\Big(\inf_{\Delta_2 \in \Lambda(\delta)} S\big(v_n^* + \sup_{\pi \in B(\delta), \Delta_1 \in \Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega + \Delta_1), \Omega + \Delta_2\big) \le c(\eta, \Omega) + \varepsilon \Big| \{W_i\}_{i=1}^n\Big) \\ + R_{n,1} + R_{n,3} < 1 - \alpha \end{array} \right\} \\
&\overset{(2)}{=} \left\{ \begin{array}{c} P\Big(\inf_{\Delta_2 \in \Lambda(\delta)} S\big(\xi + \sup_{\pi \in B(\delta), \Delta_1 \in \Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega + \Delta_1), \Omega + \Delta_2\big) \le c(\eta, \Omega) + \varepsilon\Big) \\ + R_{n,1} + R_{n,3} + R_{n,5} < 1 - \alpha \end{array} \right\} \\
&\overset{(3)}{\subseteq} \{R_{n,1} + R_{n,3} + R_{n,5} < -\lambda\},
\end{aligned}$$

where (1) holds by (A.6) for all $n \ge N$, (2) by (A.8), and (3) by (A.13). This then implies that

$$P_n(\hat{c}_n(\theta_n) > c(\eta, \Omega) + \varepsilon) \;\le\; P_n(R_{n,1} + R_{n,3} + R_{n,5} < -\lambda) \overset{(1)}{\to} 0, \tag{A.15}$$

where (1) holds by $R_{n,1} + R_{n,3} + R_{n,5} = o_{P_n}(1)$.

Finally, consider the following argument.

$$\begin{aligned}
&\{\hat{c}_n(\theta_n) < c(\eta, \Omega) - \varepsilon\} \\
&\subseteq \{P_n(Q_n^*(\theta_n) \le c(\eta, \Omega) - \varepsilon | \{W_i\}_{i=1}^n) \ge 1 - \alpha\} \\
&\overset{(1)}{\subseteq} \left\{ \begin{array}{c} P_n\Big(\sup_{\Delta_2 \in \Lambda(\delta)} S\big(v_n^* + \inf_{\pi \in B(\delta), \Delta_1 \in \Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega + \Delta_1), \Omega + \Delta_2\big) \le c(\eta, \Omega) + \varepsilon \Big| \{W_i\}_{i=1}^n\Big) \\ + R_{n,1} + R_{n,2} \ge 1 - \alpha \end{array} \right\}
\end{aligned}$$

$$\overset{(2)}{=} \left\{ \begin{array}{c} P\Big(\sup_{\Delta_2 \in \Lambda(\delta)} S\big(\xi + \inf_{\pi \in B(\delta), \Delta_1 \in \Lambda(\delta)} \varphi(\{\pi_j - \eta_j\}_{j=1}^p, \Omega + \Delta_1), \Omega + \Delta_2\big) \leq c(\eta, \Omega) + \varepsilon \Big) \\ + R_{n,1} + R_{n,2} + R_{n,4} < 1 - \alpha \end{array} \right\}$$

$$\overset{(3)}{\subseteq} \{R_{n,1} + R_{n,2} + R_{n,4} \geq \lambda\},$$

where (1) holds by (A.7) for all $n \geq N$, (2) by (A.9), and (3) by (A.14). This then implies that

$$P_n(\hat{c}_n(\theta_n) > c(\eta, \Omega) + \varepsilon) \;\leq\; P_n(R_{n,1} + R_{n,2} + R_{n,4} < -\lambda) \overset{(1)}{\to} 0, \tag{A.16}$$

where (1) holds by $R_{n,1} + R_{n,2} + R_{n,4} = o_{P_n}(1)$. Since the choice of ε was arbitrary, the proof concludes by combining (A.15) and (A.16). ■

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